

Huijie (Christopher) Qu

(424) 355-1332 • hedgequ1995@gmail.com • [linkedin.com/in/christopherqu](https://www.linkedin.com/in/christopherqu) • <https://github.com/ChristopherQu>

Senior Software Engineer with experience building large-scale streaming and batch data platforms, AI-powered data governance systems, and LLM-driven agent orchestration platforms. Specialized in distributed systems, batch/real-time data processing, ML Ops, and productionizing GenAI workflows using LangChain, LangGraph, and domain-specific LLMs.

Work Experience

Apple - Senior Software Engineer | Cupertino, CA, 03/2025 - Current

- Architected and productionized an agentic data quality platform using Python and LangGraph, leveraging a multi-agent architecture to generate 70–100 statistical and semantic validation rules per dataset from metadata, reducing manual rule authoring effort by 60%.
- Developed an LLM-powered agent orchestration platform using Python, LangGraph, and Claude SDK to automate 10+ migration of data pipelines to a domain-specific anonymized identity framework, reducing migration effort from 3 days to 5 hours through autonomous code generation, dependency analysis, and workflow planning.
- Engineered the end-to-end feature enrichment framework delivering training-ready datasets for downstream ML workloads, eliminating data prep and cutting time-to-train by 50% with async API enrichment, in-flight PII redaction, failure tagging, and retry orchestration.
- Built and owned the music library feed pipeline with Scala and Spark, turning 1B+ protobuf events/day into typed transaction data, powering music recommendation and fuzzy search.

TikTok - Data Engineer, Data Platform | Seattle, WA, 06/2024 - 03/2025

- Built a two-stage detection pipeline for Trust & Safety with Flink for real-time tag filtering and Ray for ML-driven tag relevance refinement, enabling the moderation team to catch high-priority incidents within a 30-minute end-to-end SLA.
- Re-architected the feature pipeline for video moderation models in Spark, eliminating train-serve skew via a shared feature layer, shortening model iteration cycles by 40%, and unblocking the ML team to ship 6+ model versions per quarter.
- Rebuilt the model evaluation & labeling pipelines powering Video/Live moderation QC, cutting end-to-end latency by 60% and compute cost by 30%, unblocking weekly model quality reviews and feeding ground-truth labels back into the training loop.

Amazon - Data Engineer II | Seattle, WA, 11/2021 - 06/2024

- Led development of a scalable data platform framework on AWS using Airflow, Spark, ECS, and Step Functions, enabling decentralized pipeline orchestration, standardized governance, and secure PII/PHI processing across multiple engineering teams, reducing platform maintenance cost by 80%.
- Designed a self-service data platform for BI and analytics workloads across 6+ teams, improving deployment velocity by 30% and optimizing Redshift utilization through standardized data deployment and governance practices.
- Automated migration of 400+ legacy ETL workflows to AWS via a pipeline-as-code framework, accelerating migration by 60% while standardizing CI/CD, infrastructure provisioning, and workflow orchestration.
- Transformed a PB-scale communication data pipeline from batch to real-time architecture using Elasticsearch, Firehose, and Lambda, reducing operational cost by 15%.

DOWC - Data Engineer | Parsippany, NJ, 03/2020 – Oct 2021

- Early team member who designed and built initial end-to-end data pipelines, establishing the foundation of data infrastructure.

Skills

Programming Language	Python Scala SQL Typescript JavaScript Java
LLM & AI Systems	Multi-Agent Systems Agentic Infrastructures RAG Prompt Engineering Evals Framework
Data & AI Infrastructure	Spark Flink Ray Hadoop Kafka PyTorch Hive Airflow Splunk ElasticSearch
Data Storage	Data Lakehouse Vector Database Relational (SQL) Database Non-relational (NoSQL) Database
Cloud & Dev Tools	AWS GCP Docker CI/CD Git Bash Unix Jenkins

Education

Master of Financial Engineering, University of Southern California

May 2019

Bachelor of Computational Mechanics, Xiangtan University

June 2017